

# Reality Inside and Outside the Screen

Chutisant Kerdvibulvech, Ph.D.  
Assistant Professor

ผศ.ดร.ชุตินันต์ เกติวิบูลย์เวช

Lecture for PhD Program, Department of Computer Engineering  
Faculty of Engineering, Kasetsart University

## Overview

- Background
- Technology Trends for Computer Engineering Research
- Reality Inside and Outside the Screen – Mixed/ Augmented Reality
- Journal/Conference Papers (Scopus) for Ph.D.

<http://gscm.nida.ac.th/~chutisant/KU.pdf>

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## Education



[www.chula.ac.th/](http://www.chula.ac.th/)

- B.E. (Hons) in Computer Engineering  
Chulalongkorn University



[www.keio.ac.jp](http://www.keio.ac.jp)

- MSc. / Ph.D. in Engineering, Department of Information and Computer Science  
Keio University, the oldest university in Japan

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## Academic Work Experience

- Teaching and being as a research committee in following universities, such as
  - Rangsit University, Sripatum University, PIM
  - Chulalongkorn University, Mahidol University, NIDA
- Technical Committees for many academic international conferences and journals (Scopus and Scimago)

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## Media Work Experience (1/3)

### 1. Newspapers

- IT Columnist for Dailynews newspaper, every Wednesday (page 23)



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## Media Work Experience (2/3)

### 2. Magazines

- Columnist for Manager 360° Magazine, iTech 360° Magazine, Augmented Reality Magazine, e-commerce Magazine

**ผู้จัดการ**

นิตยสารผู้จัดการ 360 องศา พฤษภาคม 2555

จาก Facebook มาสู่ GREE สร้างอาณาจักรโซเชียลมีเดียสุดโต่ง โดย ดร.บุณิสร์ เกตุบุญญะ

นิตยสารผู้จัดการ 360 องศา เมษายน 2555

2G จากอดีต มาสู่ 3G ปัจจุบัน ก้าวเข้าสู่ 4G แห่งอนาคต โดย ดร.บุณิสร์ เกตุบุญญะ



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## Media Work Experience (3/3)

### 3. Televisions

- IT News Analyst for Nation Channel TV Program



- IT News Analyst for RSU TV Channel



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## TECHNOLOGY TRENDS FOR COMPUTER ENGINEERING RESEARCH

# Digital Media

Digital media are any media that are encoded in a machine-readable format. [*Technology Brief*. University of Guelph.]

1. Digital Imagery / Digital Video
2. Web Pages / Websites
3. Digital Audio, e.g. mp3s
4. e-books
5. Social Media



[www.linkedin.com](http://www.linkedin.com)

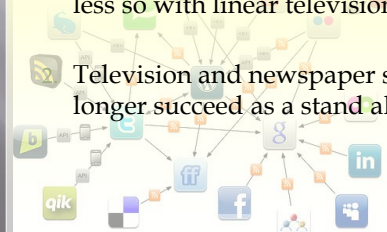
## Important Role of Digital Media

Online video has become ubiquitous, digital media consumption has become the norm.

## Facts about Digital Media

1. Viewers are heavily engaged with digital media – less so with linear television and newspaper.

Television and newspaper still matter, but it can no longer succeed as a stand alone platform.

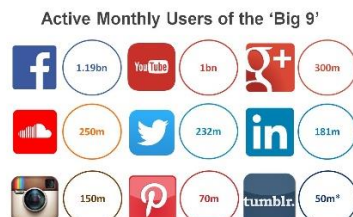


Source:  
Digital Media Trends and the Future of  
Television, Dr. Seth Geiger,  
President SmithGeiger LLC (CA, USA)

<https://www.linkedin.com/>

## Trends for Digital Media Trends for 2014

### Active monthly users of the 'big 9'



As at 15 November 2013 (multiple sources cited)

kember

Source:  
Kamber (Sydney,  
Australia)



14 Social Media Trends for 2014

Kamber.com.au | 14

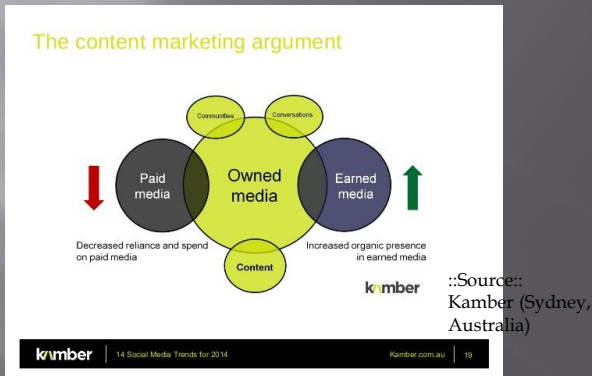
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## Mobile Phone As Home Computer

- Mobile Phone as PC
- Mobile phones are the new computer
- Designing website/social media for mobile phones is undoubtedly essential



## Paid Media / Earned Media



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Source: Kamber (Sydney, Australia)

## Creative-Digital Economy

- Placement: Its where young (and old) people are
- Price to invest: Minimal
- But..
  - Easy to build innovatively
  - Create HUGE profits



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<http://www.dailynews.co.th/>

## What's Next?

- Big Data / Cloud Computing / SEO (Search Engine Optimization) – SEM (Search Engine Marketing)
- Mobile-Oriented
- Reaching-Out-of-Computer
  - 3D-Display / 3D Television
  - MR (Mixed Reality) / AR (Augmented Reality)



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[www.3dsg.com.au](http://www.3dsg.com.au)

## 3D DISPLAY / 3D TELEVISION



## 3D Display / 3D Television



<http://www.afp.com/>

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## 3D Display / 3D Television

A free-  
floating  
image  
created by  
firing  
lasers into  
thin air



<http://www.afp.com/>

## 3D Display / 3D Television

- ❑ 2008-Current
- ❑ Hideki Kimura
- ❑ National Museum of  
Emerging Science and  
Innovation (Miraikan),  
Tokyo

2009



2014 October



<http://www.afp.com/>

## 3D Display / 3D Television



<http://www.afp.com/>

## 3D Display / 3D Television

- ❑ Miraikan is located in Odaiba, Tokyo



[www.japan-guide.com](http://www.japan-guide.com)

[www.whitehouse.gov](http://www.whitehouse.gov)

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## NEW VIEWPOINT ANALYSIS

Prof. Takeo Kanade,  
Carnegie Mellon University (CMU),  
United States

## Super Bowl

- Football fans treated to a unique new view of the action during playbacks
- CBS Television used a new technology
- Carnegie Mellon University computer vision expert, Prof. Takeo Kanade

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## Super Bowl

- ❑ “Eye Vision”
- ❑ Involves shooting multiple video images of a dynamic event, such as a football game, from multiple cameras placed at different angles
- ❑ The video streams from these cameras are combined by computer and the resulting images reach viewers in a format that will make them feel as if they are flying through the scenes they see.



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Eye Vision from Prof.  
Takeo Kanade, CMU

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## The Matrix



- ❑ "Virtualized Reality" [www.imdb.com](http://www.imdb.com)
- ❑ People will be able to customize the perspective from which they watch, for example, from that of a particular object.

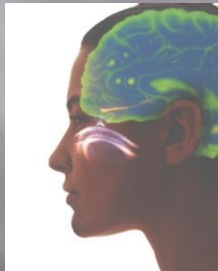
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New Viewpoint Analysis

## AUGMENTED REALITY

## What is Augmented Reality (AR)?



- ❑ A combination of a real scene viewed by a user and a virtual scene generated by a computer that augments the scene with additional information.
- ❑ Generates a composite view for the user.

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## What is the Goal of AR?

- ❑ To enhance a person's performance and perception of the world
- ❑ But, what is the ultimate



"Augmented reality in urban places: contested content and the duplicity of code." Transactions of the Institute of British Geographers, 2012

## The Ultimate Goal of AR

- Create a system such that no user CANNOT tell the difference between the real world and the virtual augmentation of it.



AR in Commerce [Geek.com](http://Geek.com)

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## Augmented Reality vs. Virtual Reality

### Augmented Reality

- System augments the real world scene
- User maintains a sense of presence in real world
- Needs a mechanism to combine virtual and real worlds

### Virtual Reality:

- Totally immersive environment
- Visual senses are under control of system

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## Miligram's Reality-Virtuality Continuum



Miligram coined the term "Augmented Virtuality" to identify systems which are mostly synthetic with some real world imagery added such as texture mapping video onto virtual objects.

Augmented Reality is closest to the real world because mainly a user is perceiving the real world with just a little computer generated data.

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## Combining the Real and Virtual Worlds

### We need:

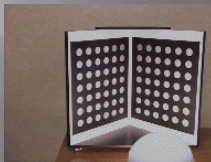
- Precise models
- Locations and optical properties of the viewer (or camera) and the display
- Calibration of all devices
- To combine all local coordinate systems centered on the devices and the objects in the scene in a global coordinate system

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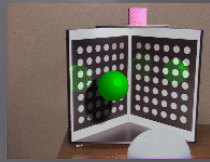


## Combining the Real and Virtual Worlds (cont)

- ❑ Register models of all 3D objects of interest with their counterparts in the scene
- ❑ Track the objects over time when the user moves and interacts with the scene



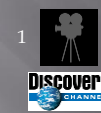
real world



real world with virtual objects and inter-reflections and virtual shading

## Mixed Reality

- ❑ Human pacman: A mobile wide-area entertainment system based on physical, social, and ubiquitous computing

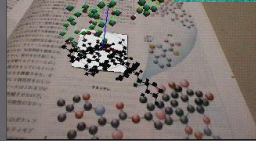


Dr. Adrian Cheok,  
National University of Singapore (NUS)  
at Singapore

Broadcast in Discovery Channel

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This image depicts and artist's conception of a mixed reality enhanced book. The reader is able to read the actual book, but simultaneously see three dimensional computer graphics co-located with, and in registration with the book.



[www.sony CSL.co.jp](http://www.sony CSL.co.jp)

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## Realistic Merging

Requires:

- ❑ Objects to behave in physically plausible manners when manipulated
- ❑ Occlusion
- ❑ Collision detection
- ❑ Shadows

**\*\*All of this requires a very detailed description of the physical scene**

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## Research Activities

- ▣ Develop methods to register the two distinct sets of images and keep them registered in real-time
  - New work in this area has started to use Computer Vision (CV) techniques
- ▣ Develop new display technologies for merging the two images

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## Performance Issues

Augmented Reality systems are expected:

- ▣ To run in real-time so that the user can move around freely in the environment
- ▣ Show a properly rendered augmented image

Therefore, two performance criteria are placed on the system:

- ▣ Update rate for generating the augmenting image
- ▣ Accuracy of the registration of the real and virtual image

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## Failures in Registration

Failures in registration due to:

- Noise
  - ▣ Position and pose of camera with respect to the real scene
  - ▣ Fluctuations of values while the system is running
- Time delays
  - ▣ In calculating the camera position
  - ▣ In calculating the correct alignment of the graphics camera

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## Applications

- ▣ Medical
- ▣ Military Training
- ▣ Hazard Detection
- ▣ Engineering Design
- ▣ Robotics and Telerobotics
- ▣ Training
- ▣ Sports
- ▣ Tourism
- ▣ Video conferencing

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## Applications (cont.)

- ▣ Entertainment, Computer Games
- ▣ Manufacturing, Maintenance, and Repair
- ▣ Consumer Design
- ▣ Audio
- ▣ Education
- ▣ Architecture
- ▣ Human welfare

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## RECENT AUGMENTED REALITY APPLICATIONS

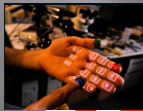
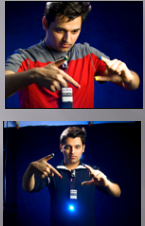
## SIXTH SENSE

## Sixth Sense

- ▣ A wearable gestural interface that augments the physical world around us with digital information
- ▣ Let us use natural hand gestures to interact with that information
- ▣ The SixthSense prototype is comprised of a pocket projector, a mirror and a camera.

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## Sixth Sense in 21<sup>st</sup> Century



Dr. Pattie Maes,  
Massachusetts Institute of Technology  
(MIT), United States

Broadcast in TED



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## Detail of Sixth Sense

- ❑ Sixth Sense allows people to use the internet without a screen or a keyboard
- ❑ It can turn any surface into a touch screen
- ❑ It is made with store-bought components
- ❑ *"The software program processes the video stream data captured by the camera and tracks the locations of the colored markers [that you wear on your fingers] using simple computer-vision techniques" ..from MIT Media Lab*

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## Detail of Sixth Sense

- ❑ The device recognizes items in stores, and then gathers and projects information about products to the user.
- ❑ It also is able to provide quick signals to let users know what the best products are based on their needs."
- ❑ Recognize articles in newspapers, retrieve the latest related stories or video from the Internet and then display them on pages for the user
- ❑ Say goodbye to your encyclopedia

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## Cost of Sixth Sense

- ❑ Only \$350 dollars, around 12,000 baht
- ❑ From an educator's stand point, this is minimum given the amount of money spent on textbooks, computer labs and LCD projectors.
- ❑ For educator's, Sixth Sense would be a way to put technology (literally) into each student's hands, because of its size and its cost.

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# AUGMENTED REALITY GUITAR

## Augmented Reality Guitar



Dr. Chutisant Kerdvibulvech,  
Keio University, Japan

Broadcast in Discovery Channel



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# MR FOR MILITARY APPLICATION

## Mixed Reality for Military Application



Dr. Adrian Cheok,  
National University of Singapore (NUS)

Broadcast in CNN



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# AUGMENTED REALITY MAGIC

## Augmented Reality Magic

- Now that an ever-accelerating cascade of eye-popping visual technology such as AR has threatened to steal some of the magic dust from old-fashioned magicians
- Along comes a pasteboard prestidigitator who folds augmented reality into



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## Augmented Reality Magic (card trick)

- Blend of technology and illusion creates the magic of tomorrow, today



Marco Tempest,  
Newmagic Communications  
Inc.,  
United States

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## Integration leads to Innovation

Traditional Methods



Conventional Frameworks



Research  
Innovation

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# HOW TO WRITE A JOURNAL PAPER

## Technical Writing



Reference: Susan Brady, Berkeley

Adapted from *The Mayfield Handbook of Technical and Scientific Writing*

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## Where do we publish?

- Conference proceedings (somewhat reviewed, progress reports) These are not all in the citation index. Why is this important?
  - ASME conferences (Are the proceedings reviewed?)
  - Special topical conferences (good for visibility)
- Journals (better reviewing, archival results)
  - Engineering journals (I. Heat Transfer (ASME), Intern. J. Heat and Mass Transfer, etc.)
  - Physics journals (Physics of Fluids, Phys. Rev., Nature Phys., Rev. Sci. Inst.)
  - Cryogenics (engineering and applied projects; Proc. of Space Cryogenics workshop)
- How to choose the right journal for your work?
  - Journal ranking (impact factor\*)
    - Scopus, Scimago, ISI
  - Journal exposure to the interested community

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## Journal Impact factor

- Why is this important?
- How is it calculated?
  - As defined by the Thomson Scientific corporation, "journal impact factor" is the average number of times that all of a journal's articles published in the two years previous to a chosen year (ex: 2004-2005) have been cited in that chosen year (ex: 2006).
- Most prestigious journals have high impact factors: e.g. Nature (27), Science (22)
- Engineering journals generally do not have high impact factors (Cryogenics ~ 1)
- Playing the game... (downloading, citing yourself...)

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## What

- Length of an article is often determined by the publication requirements.
  - Cryogenics has only suggested manuscript lengths (10 pages for Tech note; 20 pages for research article)
  - Phys. Of Fluids has limits (4 pages for Letter)
  - Don't try to cram too much into a 4 page paper
- Headings form the outline of the manuscript
  - Most people do not read an entire article so you need to make it easy to find what they are looking for.

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## Tone & Style

"This is not classic literature"

- Use direct declarative sentence structure. Avoid lengthy and indirect sentences. (example)

"Having acquired the components and assembled them in the laboratory, the system was built."

"The system was built and assembled using components from industry."
- Use professional, non-combative language:

"Van Sciver, et al did not consider..." is OK

"Van Sciver, et al totally overlooked (or are out of their mind)..." is not.
- Keep in mind your audience. Most readers are interested in the What, Why and How of your work. Don't make it hard to find.
- Don't weigh your paper down with minutia.
- Don't repeat yourself; don't repeat yourself; don't...

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## Formatting

- Most journals have on-line format instructions. Some cases include a manuscript template
  - Sure way to annoy a conference paper referee; do a poor job of laying out your paper
  - Many conference proceedings are going to do a format check before the technical editor even sees the manuscript
- MS word is acceptable for Elsevier journals, but others (Springer) expect publications to be submitted in Latex format ☹
- Initial submission format often not critical since most articles are sent back to author for revision.
- Pay particular attention to reference format which is different depending on journal

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## Parts of a Manuscript

- Title: "...should summarize the main idea of the paper simply and with style". Should be ~ 10 to 12 words. Inverse relationship between title length and significance of article:
  - "Theory of Superconductivity" by Bardeen, Cooper and Schrieffer
  - "3-D Numerical Analysis for Heat Transfer from a Flat Plate in a Duct with Contractions Filled with Pressurized He II"
- Authorship: This is a topic that causes some concern. Generally, co-authorship should be limited to those who have materially contributed to the research and preparation of the manuscript. Subtle issue.

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## The Abstract

- “...a brief, comprehensive summary of the contents of the article.” “...can be the most important paragraph in the article.”
- Abstracts are published separately from articles in on-line indices, so make it clear.
- An abstract should be
  - Accurate: should reflect the content of the paper.
  - Self-contained: avoid abbreviations, acronyms, define unique terms. Avoid references or equations. Summarize conclusions.
  - Concise and specific: Be as brief as possible, yet convey the information. 5% of article or 500 words at most
  - Compare abstract content to outline of paper. (Reference 1)
- Examples of good and bad abstracts

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## Introduction/Background/Literature Discussion

- Introduction usually contains background information
- Avoid repeating the contents of the Abstract
- Is needed to place your work in context.
- Most citations occur here. Avoid critical statements. Keep in mind that most potential referees are authors of similar work.
- Typically should not exceed about 25% of total length.
- Often the Introduction is the most difficult section to write. Probably not the best section to start writing this section first.

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## Experiment or Analysis discussion

- Describe the apparatus and method used to obtain the data.
  - Avoid too much detail (part numbers, model numbers, unnecessary dimensions)
  - Experimental schematic is more valuable than a photo of the outside of the apparatus
- Reference to other similar experiments:
  - “this apparatus, which was originally developed for liquid oxygen viscosity measurements, was modified...”
  - “our design is similar to that of Diller<sup>6</sup>, et al...”
- Describe the data collection and analysis.
  - If appropriate include error discussion, but keep it brief!
  - A possible sentence might be: “We recorded data over a range of temperatures between 1.7 and 2.1 K and pressures to 1 MPa.”
- Present only that necessary to understand the experiment, but be complete.
- It's OK to repeat some things published elsewhere if it helps the reader avoid looking up another reference.
- This is often the easiest section to write and thus might be a good place to start

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## Results and Discussion

- This section typically contains tables and graphs of all the data and analysis comparison.
- Should be compact. Don't attempt to show everything. A sample of the data compared to a general correlation is good. 15 plots of data for different operational conditions w/o explanation is not.
- Rough rule of thumb: No more than ~ one figure per page of text
- Do not duplicate data in tables that are adequately presented in graphs (precision data is better in tables)
- Compare your results with other similar experiments, if appropriate. In this case, be sure to use complete references.

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## Summary/Conclusion

- ❑ Keep this section short!
- ❑ State the most important findings and how your work has advanced the field.
- ❑ Possibly comment on what additional work would be beneficial or is planned.

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## Other Stuff

- ❑ Acknowledgment: indicate thanks to those who have helped with the work, but are not co-authors. List funding source:
  - Thanks to Scott Maier for technical assistance; SVS acknowledges helpful suggestions from Dr. S. Fuzier
  - Work supported by the US Department of Energy under grant DE-FG-02-96ER-40952
  - This research has been supported by NASA through the Research Initiative for Florida Universities under the grant NAG3-2751
- ❑ References: In order cited in the paper. Be complete here. This is how citation indices are constructed. Worst thing is to overlook an important reference from one of the referees. Formats depend on journals.

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## Writing your manuscript

- ❑ Choose your time for writing so that there are few interruptions (examples)
- ❑ Set aside a block of time (min ~ 2 to 3 hours) preferably every day so that you maintain continuity
- ❑ Make an outline: with annotations and references. Build the text within the outline
- ❑ Set a goal for each time period. e.g. finish a section.
- ❑ Keep your effort up until you produce a draft.
- ❑ If writing a PhD dissertation remember the “P” stands for “perseverance”

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## Strategy to getting started

- ❑ Make that annotated outline
- ❑ Possible approach #1: collect data and decide on graphics, equations. Build the outline around the data to be displayed.
- ❑ Conference paper approach: prepare the talk first and use the Power Point as the outline of your paper. Give the talk and flesh out your logic, discussion that you can later write.
- ❑ Proof read often
- ❑ Have a colleague proof read what you have written, particularly if English is not your native language

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## Time

- ▣ “is a wasting...”
- ▣ Don’t wait to the last minute, so get started now.
- ▣ ..and don’t stop until you have a first draft
- ▣ No one said this would be easy, but it **IS** important

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